

Stage1 Gears — ISO 6336 Bending & Contact Stress Check

Tooth Counts & Ratio Check

Pinion tooth count, z_1	23
Gear tooth count, z_2	36
Gear ratio, $u = z_2/z_1$	1.57
Target stage ratio (from Inputs)	1.58

Geometry

Normal module, m_n	2.50
Face width, b (trial - iterate to hit target SF)	25.0
Pinion pitch diameter, $d_1 = m_n \cdot z_1 / \cos(\beta)$	57.50
Gear pitch diameter, $d_2 = m_n \cdot z_2 / \cos(\beta)$	90.00
Center distance, $a = (d_1 + d_2) / 2$	73.75
Face width ratio, b/d_1	0.43

Loads

Input speed at this pinion	11500.0
Input torque at this pinion	63.0
Pitch line velocity, $v = \pi \cdot d_1 \cdot n / 60000$	34.62
Tangential load, $F_t = 2000 \cdot T / d_1$	2191.3

Transverse Contact Ratio — CALCULATED (real involute geometry)

Pinion base radius, $rb_1 = (m_n \cdot z_1 / 2) \cdot \cos(\alpha_n)$	27.02
Gear base radius, $rb_2 = (m_n \cdot z_2 / 2) \cdot \cos(\alpha_n)$	42.29
Pinion tip radius, $ra_1 = m_n \cdot z_1 / 2 + m_n$ (std addendum)	31.25
Gear tip radius, $ra_2 = m_n \cdot z_2 / 2 + m_n$ (std addendum)	47.50
Transverse contact ratio, ϵ_{α}	1.642

Load & Application Factors

Application factor, K_A (guided input - see README)	1.50
Dynamic factor, K_v — CALCULATED (AGMA-equivalent closed form, see README)	
Quality number, Q_v (from Inputs)	6.0
$B = 0.25 \cdot (12 - Q_v)^{2/3}$	0.825
$A = 50 + 56 \cdot (1 - B)$	59.77
v in m/min = $v(m/s) \cdot 60$	2077.4
$K_v = ((A + \sqrt{3.28 \cdot v_{m/min}}) / A)^B$	2.046
Face load factor - contact, K_{Hb} (guided - see README, b/d_1 shown above)	1.25
Face load factor - bending, K_{Fb} (guided, ~ K_{Hb} but slightly lower)	1.22
Transverse load factor - contact, K_{Ha} (guided, Grade 8 typical)	1.10
Transverse load factor - bending, K_{Fa} (guided, Grade 8 typical)	1.10

ISO 6336-3 Bending Geometry Factors

Contact ratio factor, $Y_{\epsilon} = 0.25 + 0.75 / \epsilon_{\alpha}$ — CALCULATED	0.707
Helix angle factor, $Y_{\beta} = 1 - \epsilon_{\beta} \cdot (\beta / 120)$ — CALCULATED ($\epsilon_{\beta} = 1$, spur)	1.000
Tooth form factor, Y_{Fa} (guided - chart value for $z \sim 20$, see README)	2.85

Stress correction factor, YSa (guided - chart value for z~20)	1.55
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ISO 6336-2 Contact Geometry Factors

Zone factor, ZH — CALCULATED	2.495
Elasticity factor, ZE (steel-steel, fixed) — CALCULATED (constant)	189.8
Contact ratio factor, Zeps = $\text{SQRT}((4-\epsilon_{\alpha})/3)$ — CALCULATED (spur)	0.887
Helix angle factor, Zbeta = $\text{SQRT}(\cos(\beta))$ — CALCULATED (=1, spur)	1.000

Material & Life Factors

Bending endurance limit, sigma_Flim (from Inputs)	430
Contact endurance limit, sigma_Hlim (from Inputs)	1500
Stress correction factor, YST (standard reference gear)	2.00
Life factor - bending, YNT (1.0 = finite/reference life; >1 if very low cycles)	1.10
Life factor - contact, ZNT	1.10
Other bending factors lumped, YdeltaT*YRelT*YX	1.00
Other contact factors lumped, ZL*ZV*ZR*ZW*ZX	1.00
Minimum required bending safety factor, SFmin	1.00
Minimum required contact safety factor, SHmin	1.00

Bending Stress Check (ISO 6336-3)

$\sigma_F = F_t \cdot K_A \cdot K_v \cdot K_{Fb} \cdot K_{Fa} \cdot Y_{Fa} \cdot Y_{Sa} \cdot Y_{\beta} \cdot Y_{\epsilon} / (b \cdot m_n)$	450.9
$\sigma_{FP} = \sigma_{Flim} \cdot Y_{ST} \cdot Y_{NT} \cdot Y_{other} / SF_{min}$	946.0
Bending safety factor, SF = σ_{FP} / σ_F	2.10

Contact Stress Check (ISO 6336-2)

$\sigma_{H0} = Z_E \cdot Z_H \cdot Z_{\epsilon} \cdot Z_{\beta} \cdot \text{SQRT}(F_t \cdot (u+1) / (b \cdot d_1 \cdot u))$	663.5
$\sigma_H = \sigma_{H0} \cdot \text{SQRT}(K_A \cdot K_v \cdot K_{Hb} \cdot K_{Ha})$	1363.1
$\sigma_{HP} = \sigma_{Hlim} \cdot Z_{NT} \cdot Z_{other} / SH_{min}$	1650.0
Contact safety factor, SH = σ_{HP} / σ_H	1.21